

admits a representation of the form:

$$f(\mathbf{x}) = \sum_{i=1}^m \alpha_i k(\mathbf{x}_i, \mathbf{x}). \quad (3.144)$$

3.8 Exercises

3.1 Given $\mathbf{X}\mathbf{w} = \mathbf{y}$ where $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 5 & 6 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

- (a) What kind of system is this? (even-, over- or under-determined?)
- (b) Is $\mathbf{X}$ invertible? Why?
- (c) Solve for $\mathbf{w}$ if it is solvable

3.2 The linear system given by $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is over-determined and thus has no exact solution. True or False?

3.3 Consider the following system:

$$\begin{aligned} w_1 + w_2 &= 0 \\ w_1 - w_2 + w_3 - w_4 &= 1 \\ w_3 + w_4 &= 2. \end{aligned}$$

What kind of system is this? Can the system be solved? Select one or more from below:

- (a) The system is even-determined and can be solved with an exact solution
- (b) The system is over-determined and can be solved with an approximated solution
- (c) The system is over-determined and it does not have an exact solution
- (d) The system is under-determined and can be solved with an exact least norm solution
- (e) The system is under-determined and it has infinite number of solutions

3.4 Identify those polynomial models shown below which include at least a second-order term:

- (a) $1 + x_1 + x_2 + x_1x_2 - x_1^2 - x_2^2$
- (b) $1 + x_1 + x_2 + x_3 + x_4 + x_5$
- (c) $1 + x_1 + x_2 + x_1x_2$
- (d) $1 + x_1 + x_2 + x_1(x_1 - x_2)$
- (e) None of these

3.5 Given the input feature x and the corresponding response y as follows:

x	-10	-7	-3	-1	4	8
y	10	11	8	6	3	4

- (a) Perform a third-order polynomial regression and plot the result of line fitting
- (b) Given a test point $x = -5$ predict y using the learned polynomial model
- 3.6 This question is also about polynomial models. Which of the following models has discriminative feature for the data given by

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} +1 \\ +1 \end{bmatrix}, \begin{bmatrix} -1 \\ +1 \end{bmatrix}, \begin{bmatrix} +1 \\ -1 \end{bmatrix}$$

with corresponding target values $y \in \{+1, +1, -1, -1\}$?

- (a) $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2$
- (b) $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2 + w_3x_1x_2$
- (c) $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2 + w_3x_1^2$
- (d) $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2 + w_3x_2^2$
- (e) $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2$
- (e) None of these.
- 3.7 This question is also about multivariate polynomials.
- (a) Write down the expression for a third-order polynomial model having a three-dimensional input
- (b) Write down the $\mathbf{P}$ matrix for this polynomial given
- $$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$
- (c) Given $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, can a unique solution be obtained for this under-determined system of $\mathbf{P}$ matrix?
- 3.8 This question is about the reduced multivariate polynomials.

- (a) Write down the expression for a third-order RM model having a three-dimensional input
- (b) Write down the $\mathbf{P}$ matrix for this polynomial given
- $$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$
- (c) Given $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, can a unique solution be obtained for this under-determined system of $\mathbf{P}$ matrix?
- 3.9 The two-layer perceptron network with linear output layer can be written in two ways, namely,

$$\mathbf{g}(\mathbf{W}_o, \mathbf{W}_h, \mathbf{x}) = \mathbf{W}_o L(\mathbf{W}_h \mathbf{x}), \quad \mathbf{W}_h \in \mathbb{R}^{h \times d}, \quad \mathbf{W}_o \in \mathbb{R}^{o \times h},$$

and

$$\mathbf{g}(\mathbf{W}_o, \mathbf{W}_h, \mathbf{x}) = L(\mathbf{x}^T \mathbf{W}_h) \mathbf{W}_o, \quad \mathbf{W}_h \in \mathbb{R}^{d \times h}, \quad \mathbf{W}_o \in \mathbb{R}^{h \times o},$$

where $L(\cdot) = \max(0, \cdot)$ for $\mathbf{x} \in \mathbb{R}^{d \times 1}$. Let's use the second form for our computation and assume $\mathbf{W}_h$ has been fixed at

$$\mathbf{W}_h = \begin{bmatrix} 2 & 2 & 3 \\ 3 & 0 & 2 \end{bmatrix}.$$

Suppose you are given several training samples $\mathbf{x}$ with corresponding target y :

$$\mathbf{x} : \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}, \quad y : \{0, 0, 1, 1\}.$$

Can the output weights $\mathbf{W}_o$ be solved deterministically?

- 3.10 Code a function to build a regressor matrix for the Reduced Multivariate Polynomials Reciprocal Sigmoid features with MATLAB/Octave.